WIPRO NGA Program – C P Java

Capstone Project Presentation – 16th Oct 2024

Project Title Here – Healthy Meals

Presented by – SRIMUGAMBIGAI G &
MAYUR VILAS CHAUDHARI

Problem Statement

Content:

- Busy professional and students struggle to maintain healthy eating habits.
- Lack of time to prepare meals leads to unhealthy food choices.
- Existing meal delivery services don’t focus on nutritional value or provide an intuitive ordering experience.
- Many existing platforms offer limited options for healthy meals, often leading to menu fatigue and a lack of motivation to stay consistent with healthy eating.
- Healthy meal services may not have enough local meal providers, reducing availability in smaller towns or suburban areas.
- Consumers struggle to communicate with providers regarding meal adjustments or leave detailed feedback on the service, preventing improvements.

learn..evolve

2024 - RPS Consulting all rights reserved

Need for the project/Pain points

Why Healthy-Meals?

- Rising demand for healthy, convenient meals.
- Limited options for users seeking balanced, nutritious meals tailored to their preferences.
- Current solutions lack personalized meal plans and real-time availability updates.
- Offers meals for various dietary needs, including vegetarian, vegan, gluten-free, keto, and low-carb diets, providing greater variety than competitors.
- Our service strives to make healthy eating affordable and accessible by offering competitively priced meals without sacrificing quality.
- Healthy-Meals prioritizes partnerships with local providers and chefs who source fresh ingredients, promoting both nutrition and sustainability.

learn..evolve

2024 - RPS Consulting all rights reserved

Solution Design

Our Approach

- Provide a platform for users to order healthy, customizable meals.
- Implementing secure login mechanisms (OAuth, JWT, etc.) for both meal providers and customers ensures data protection and privacy while maintaining smooth user access. Integration of payment gateway (PayPal) for seamless transactions.
- Seamless and secure payment transactions through trusted gateway like PayPal. Users can pay quickly and confidently, knowing their financial information is protected.
- The app features live updates on meal availability, ensuring users can view what's available before placing orders, reducing cancellations and delays.
- Real-time meal availability checks and feedback system.

RPS learn..evolve
2024 - RPS Consulting all rights reserved

Microservices

- **Authentication Service**: Manages secure login and role-based access control for Admin, Customer, and Provider accounts. Implements modern authentication methods (OAuth, JWT tokens) for secure user management.

- **MealOrder Service**: Handles the creation, modification, and removal of meal options by providers. Updates meal availability based on stock and provider input, ensuring users see current options. Supports meal pricing updates, discounts, and promotional offers, ensuring users receive up-to-date costs. Manages the entire order lifecycle, from order placement to tracking and delivery status updates. Integrates with delivery partners to streamline logistics and provide real-time updates to users.

- **PaymentNotification Service**: Seamless integration with payment gateways PayPal for secure transactions. supports credit card, debit card, and wallet payments, providing flexibility for users.

- **Review & Rating Dashboard**: Allows customers to rate and review meals and services, providing essential feedback to providers and the platform. Admins and providers can view aggregated feedback and ratings, helping them track meal performance and user satisfaction.

learn..evolve

2024 - RPS Consulting all rights reserved

Discovery server/API Gateway

Centralized Communication

- **Eureka Discovery Server**: Microservices (Authentication, MealOrder, PaymentNotification, ReviewRatingDashboard) automatically register themselves with the Eureka Discovery Server upon startup, making them discoverable to other services. Supports dynamic service registration and deregistration, enabling the app to scale efficiently based on demand without manual intervention.

- **API Gateway**: Acts as a single entry point for all client requests, forwarding them to the appropriate microservice (Authentication, MealOrder, PaymentNotification, ReviewRatingDashboard) based on the request type. Handles authentication, rate-limiting, and other security measures before passing requests to microservices, ensuring a secure system.

learn..evolve

2024 - RPS Consulting all rights reserved

Dockerization

Containerization with Docker

- Each microservice (Authentication, MealOrder, PaymentNotifation, ReviewRating) is containerized using Docker, ensuring that they run in isolated, consistent environments, independent of the host machine's setup.
- Docker containers ensure that all services run in the same environment across different platforms (development, testing, production), eliminating issues caused by environment differences.
- Docker Compose simplifies the process of managing multiple microservices locally by defining and running them together with a single configuration file.
- Automatically handles networking between containers, ensuring seamless communication between services without the need for manual setup.
- Docker Compose enables easy configuration for different environments (development, staging, production) by specifying unique settings for each.

RPS learn..evolve
2024 - RPS Consulting all rights reserved

Tech Stack

Technology Stack

- **Backend**: A robust Java framework that simplifies the development of scalable and production-ready microservices, we have used spring boot microservices and JWT for secure authentication and to store all the data , used MYSQL database.

- **Frontend**: Angular for powerful front-end web framework developed by Google, used for building dynamic, single-page applications.

- **DevOps**: Docker which enables containerization of microservices. Kubernetes, : A powerful container orchestration platform that automates the deployment, scaling, and management of containerized applications. Prometheus/Grafana for monitoring, ELK stack is used to aggregate, store, and visualize logs from various services, simplifying debugging and performance analysis across the entire application.

RPS learn..evolve

2024 - RPS Consulting all rights reserved

Backlog

Project Backlog

- **Phase 5**: Testing, Monitoring, and Kubernetes Deployment.

learn..evolve

2024 - RPS Consulting all rights reserved

Conclusion

Final Thoughts

- Healthy-Meals is meticulously crafted to make healthy eating not only accessible but also convenient for our users. By focusing on customization and user-friendly features, we aim to empower individuals to make better dietary choices effortlessly.

- The implementation of a microservices architecture guarantees scalability, reliability, and ease of integration with other services, allowing us to adapt swiftly to changing market demands and user needs.

- Looking ahead, our future plans include expanding our meal options to cater to a wider variety of dietary preferences and introducing AI-powered meal recommendations. This will enhance user engagement by providing personalized suggestions, making the Healthy-Meals experience even more tailored and relevant.

RPS learn..evolve

2024 - RPS Consulting all rights reserved

Thank you!

2024 - RPS Consulting all rights reserved